Skills Summary

Taylor Polynomials and Remainder Bounds

Use this reference while completing your worksheet or when studying.

A Taylor polynomial answers “what value?”; a remainder bound answers “how wrong could that be?” — and the second is what an exam grades.

1. Building a Taylor polynomial from derivative values

Rule: $T_n(x) = f(c) + f'(c)(x-c) + \dfrac{f''(c)}{2!}(x-c)^2 + \cdots + \dfrac{f^{(n)}(c)}{n!}(x-c)^n$.

The kth derivative *value* is the numerator, $k!$ the denominator, and the power of $(x-c)$ matches k; Maclaurin is the case $c = 0$.

(!) The factorial is what forces $T_n^{(k)}(c) = f^{(k)}(c)$ — drop it and the polynomial no longer matches the function at all.

Example: $f(1) = 2$, $f'(1) = 6$, $f''(1) = -4$. Approximate $f(1.5)$ with T_2 about $c = 1$.
Step 1. Only three derivative values are given, so T_2 is the best available and every term is built on the increment $x - 1$.
Step 2. $T_2(x) = 2 + 6(x-1) - \dfrac{4}{2}(x-1)^2$, and at $x = 1.5$ the increment is 0.5: $2 + 3 - 2(0.25)$.
4.5

Try it: $g(3) = 1$, $g'(3) = -2$, $g''(3) = 6$. Approximate $g(3.5)$ with T_2 about $c = 3$.

check: **0.75**

2. Building one from a known Maclaurin series

Memorise four: $e^u = \sum \dfrac{u^n}{n!}$; $\quad \sin u = u - \dfrac{u^3}{3!} + \dfrac{u^5}{5!} - \cdots$; $\quad \cos u = 1 - \dfrac{u^2}{2!} + \dfrac{u^4}{4!} - \cdots$; $\ln(1+u) = u - \dfrac{u^2}{2} + \dfrac{u^3}{3} - \cdots$

Then substitute $u = -x^2$, $u = 2x$, $u = x^3$, whatever the problem needs, and collect up to the degree asked for — differentiating e^{-x^2} six times gives the same answer for ten times the work.

(!) Substitution changes which *degree* each term reaches: with $u = -x^2$, the $n = 3$ term of e^u is already degree 6.

Example: Use the degree-4 Maclaurin polynomial for $\cos x$ to approximate $\cos\left(\frac{1}{2}\right)$.
Step 1. Cosine’s series is known, so no differentiating is needed — just stop it at degree 4 and substitute.
Step 2. $1 - \dfrac{x^2}{2} + \dfrac{x^4}{24}$ at $x = \frac{1}{2}$ is $1 - \dfrac{1}{8} + \dfrac{1}{384} = \dfrac{384 - 48 + 1}{384}$.
$\mathbf{\dfrac{337}{384}}$

Try it: Use the degree-4 Maclaurin polynomial for e^{-x} to approximate e^{-1}.

check: $\mathbf{\frac{3}{8}}$

3. Bounding the remainder

Lagrange: $|R_n(x)| \le \dfrac{M\,|x-c|^{n+1}}{(n+1)!}$, where $\left|f^{(n+1)}(z)\right| \le M$ for *every* z between c and x. Both the exponent and the factorial use $n+1$ — using n is the most common error in the topic.

Alternating series: if the series alternates with terms decreasing to 0, the error is smaller than the **first omitted term** — and its sign says whether the estimate is high or low.

Finding M: bound the $(n+1)$st derivative on the interval — take $M = e^b$ or larger for e^x on $[0, b]$, and $M = 1$ for sin or cos.

Example: $|f'''(z)| \le 6$ on $[0, 0.3]$. Bound the error when T_2 about $c = 0$ approximates $f(0.3)$.
Step 1. The polynomial has degree $n = 2$, so the bound needs the third derivative and the third power — that is what $n+1$ means.
Step 2. $\dfrac{6 \cdot |0.3|^3}{3!} = \dfrac{6 \cdot 27/1000}{6}$.

$\mathbf{\dfrac{27}{1000}}$

Try it: $\left|f^{(4)}(z)\right| \le 24$ on $[1, 1.25]$. Bound the error when T_3 about $c = 1$ approximates $f(1.25)$.

check: $\mathbf{\frac{1}{256}}$

4. Using a polynomial to settle a limit

Replace each function by its Maclaurin series, cancel what the numerator subtracts away, and divide term by term: **the first surviving term is the limit**.

(!) Keep enough terms — truncating e^x at $1 + x$ makes $e^x - 1 - x$ look like exactly 0.

Example: Evaluate $\displaystyle\lim_{x\to 0} \frac{\sin x - x}{x^3}$ using series.
Step 1. Subtracting x kills the leading term of $\sin x$, so the limit is decided by the first term that survives.
Step 2. $\sin x - x = -\dfrac{x^3}{6} + \dfrac{x^5}{120} - \cdots$, so dividing by x^3 leaves $-\dfrac{1}{6} + \dfrac{x^2}{120} - \cdots$

$\mathbf{-\dfrac{1}{6}}$

Try it: Evaluate $\displaystyle\lim_{x\to 0} \frac{1 - \cos x}{x^2}$ using series.

check: $\mathbf{\frac{1}{2}}$